

## Kamran (Kam) Amini

Senior Systems & Solutions Architect

Independent Consultant

- ✓ CTO Advisory & Strategic Decision Making
- ✓ Systems & Solutions Architecture
- ✓ Blockchain & AI Adoption
- ✓ Data & Network Security

Author of Books & Articles on Software Engineering

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## SUMMARY

With 15+ years of experience across software development and systems architecture, I have worked in startups, large corporations, and consulting engagements as an engineer, Tech Lead, CTO, board member, and founder, giving me first-hand insight into the technical, financial, and organizational dynamics of software firms at every scale. I combine hands-on engineering with architectural leadership, coding critical components myself so my decisions stay informed by implementation reality. My core expertise lies in systems-level languages (C, C++, Golang, Rust), plus DevOps on Kubernetes and AWS/GCP. In blockchain and fintech, I architected tokenized-securities infrastructure (on-chain exchange and auction engines with atomic settlement, a central securities depository, brokerage and custody) plus ERC-20/ERC-721 token and NFT platforms. I now build AI platforms end to end (self-hosted LLMs, vision and diffusion models, LoRA training) and work daily with AI coding agents (Claude, Codex, Gemini). My background is in data and network security, and I treat security and privacy as first-class citizens in my architecture and consulting. I bring this combined perspective to the systems architecture, CTO-level strategy, and blockchain/AI consulting I provide today.

## CAREER

### DeKrypt B.V.

**Founder, CEO & CTO, Principal Architect & Consultant**

*March 2024 - Present (alongside employment, with consent)*

The Netherlands, Amsterdam

- **Consultancy:** Delivering all part-time architecture engagements listed in this CV through the firm.
- **SoulForge (own startup):** Building an AI persona platform: self-hosted LLM and vision models, FLUX image generation, per-persona LoRA training, face-identity conditioning, elastic GPU nodes.
- **FamBuzz (Buzzar) & Bo2Bo2 (own startups):** Building a private group social app and a mobile fitness game, with Golang backends on Kubernetes and Android/iOS clients.
- **AI-Driven Development:** Delivering with AI coding agents (Claude, Codex, Gemini), keeping accountability with engineers.
- **FinTech & Distributed Ledger:** Architecting exchange, settlement, custody, and treasury systems for Vero Labs, Assetera, and Metapolitan.
- **Writing & Research:** Authoring **The Software Ecology** (to be published), alongside essays and academic work for peer review.

### Vero Labs Inc.

**Senior Blockchain Architect (Part-time consultancy)**

*April 2024 - September 2026*

United States, New York

#### ↳ Tokenise Ltd

**Senior Blockchain Architect (Part-time consultancy)**

United Kingdom, London

#### ↳ VeroPulse GmbH

**Chief Software Architect & Tech Lead (Part-time)**

Austria, Vienna - led a team of 6

- **Agora Stock Exchange Engine:** Architected the on-chain orderbook and settlement engine, with atomic settlement finality enforced between the orderbook and treasury contracts.
- **Tokenise CSD:** Designed the Central Securities Depository platform responsible for on-chain issuance, custody, and settlement of tokenized securities.
- **Tokenise Broker:** Architected the brokerage platform connecting investors to the Agora Engine for order routing and execution.
- **Vero Android App:** Directed the architecture of the primary and secondary market mobile client, integrated end-to-end with the Agora Engine.
- **Platform & Infrastructure:** Designed the off-chain microservices layer in Golang, deployed on Kubernetes (AWS EKS) for scalable, highly-available trading workloads.
- **AI-Assisted Engineering:** Introduced AI-assisted coding workflows (LLM copilots, automated code review) across the teams to accelerate delivery and improve code quality.
- Golang - Kubernetes (EKS) - AWS - Solidity - EVM chains - PostgreSQL - Docker - gRPC

### Assetera GmbH

**Senior Blockchain Architect & Tech Lead (Part-time)**

*April 2024 - April 2026*

Austria, Vienna

- **Agora Auction Room:** Architected the on-chain auction mechanics, with atomic settlement finality enforced against the on-chain treasury.
- **Tokenization as a Service:** Architected the on-chain asset tokenization framework, covering issuance, custody, and transfer of tokenized assets.
- **Platform & Infrastructure:** Designed the off-chain Golang services, deployed on Kubernetes (AWS EKS).
- Golang - Solidity - EVM chains - Kubernetes (EKS) - AWS

### Digital First GmbH

**Senior Blockchain Architect (Part-time consultancy)**

*December 2023 - September 2026*

Austria, Vienna

- **TheKiss NFT Tokenization Platform:** Designed the NFT minting architecture and smart contract framework underpinning the platform's digital collectibles.
- **ARTEO Token:** Architected the ERC-20 token economics and smart contract suite for the art-backed digital asset.
- **QLINDO Token:** Led smart contract architecture and security review for the project's token launch.
- **Hergé NFTs:** Architected the ERC-721 minting and royalty framework for the licensed NFT collection.
- Solidity - EVM chains - ERC-20 - ERC-721 - Hardhat - OpenZeppelin

### Metapolitan Ltd.

**Chief Blockchain Architect (Part-time consultancy)**

*January 2025 - March 2026*

United Kingdom, London

- **Distributed Treasury System:** Architected a multi-signature, on-chain treasury platform for fintech clients, enabling secure custody, automated settlement, and policy-driven fund controls across multiple blockchain networks.
- Solidity - Gnosis Safe (multi-sig) - EVM chains - web3.js - Hardhat - Node.js

### ActiveVideo B.V. (April 2019 - February 2023)

↳ **Commscope Inc. (March 2023 - March 2024)**

↳ **ARRIS B.V. (April 2024 - December 2025)**

↳ **Vistance Networks B.V. (January 2026 - Present)**

**Senior Software Engineer**

The Netherlands, Hilversum - one company, many reorganizations

- **CloudTV (Video Stitching Engine):** Cloud renderer/streamer delivering UI and audio/video to set-top boxes.

- AppCloud: Containerized Android streaming platform with a Kubernetes (AWS EKS) backend.
- Multiview Engine: Cloud streamer broadcasting multiple channels at once to set-top boxes and TVs.
- Tech Council & Security Guild: Prioritized tech debt and enforced security standards across teams.
- C++17 (g++, clang) - Golang - Java and Kotlin - Python - AWS - Kubernetes (EKS) - S3 - EFS - EBS - Ceph - Envoy

#### TomTom B.V.

##### Senior Software Engineer

September 2017 - March 2019

The Netherlands, Amsterdam

- Expert Software Consultant from Modis B.V.
- NavKit Free Text Search Engine, NavKit Worker REST Services, POI Info Services
- C++11 (g++, clang) - Java SE - JUnit - Spring MVC - JAXB - Spring REST - gdb - Valgrind - Clang ASAN & TSAN - STL & Boost - Google Test - Protobuf - Thrift - Python - Jenkins - QuickBuild - Docker - JIRA - Fisheye - Confluence - Bitbucket Stash - AWS/Azure based cloud

#### Jeppesen Systems (a Boeing subsidiary)

##### Senior Software Engineer

February 2017 - August 2017

Sweden, Gothenburg

- Manpower Application
- C++11 (g++) - Python - C#.NET - gdb - Valgrind (callgrind, massif) - PDB - STL & Boost - Google Test - XMLRPC - Jenkins - JIRA - Fisheye - Confluence

#### Adro Co.

##### Senior Software Architect, Principal C++ Engineer

June 2016 - January 2017

Iran, Tehran

- Realtime Bidding (RTB) Engine
- C++ (g++) - Python - gdb - Valgrind - STL & Boost - libuv - Google Test - Protobuf - Jenkins - Travis CI - Docker - JIRA - Fisheye

#### Naeem Co.

##### Chief Technical Officer, Chief Software Architect

April 2015 - October 2016

Iran, Tehran

- Projects: ePassport Issuance System - Face Recognition System, Biometric Identification System
- Cryptography: OpenSSL - PKCS (1, 7, 10, 11, 12) - ICAO MRTD standards - HSM - Key Management - PKCS11 Tokens
- Stack: C (gcc) - C++ (g++) - Java SE - Java EE - Spring MVC - Wildfly - Tomcat - Python - gdb - Valgrind - STL & Boost - wxWidgets - Qt - Google Test - Jenkins - Docker - vSphere - Proxmox - Gluster - Cassandra - Redmine - GitLab

#### Sharif CERT Center

##### Software Engineer

December 2011 - June 2013

Iran, Tehran

- Botnet Detection System - Sinkhole - DDNS as a FreeBSD module
- C++ (g++) - Java SE - Perl - gdb - JConsole - Valgrind - STL & Boost - Swing - Redmine

## TECHNICAL SKILLS

- **Languages & Architecture**: C++ (14/17/20), C, Golang, Rust, Python, Java, JS - Distributed systems (CAP, replication, leader election, CQRS, event sourcing, message queues, streaming, async/event loops)
- **Cloud & DevOps**: Kubernetes (kubect1, Helm, Kustomize, PSA) - AWS & GCP (EKS/GKE, EC2/GCE, S3, VPC, IAM) - Docker - PostgreSQL, MySQL, MongoDB, Redis
- **Blockchain**: Solidity (EVM chains), Solana, web3.js/ethers.js, Hardhat - Distributed ledgers, smart contract architecture
- **Security & Cryptography**: OpenSSL, PKI, symmetric/asymmetric algos, hashing/MAC - Protocols & standards (gRPC, Protobuf, REST, PKCS, RFCs) - Networking (TCP/IP, Wireshark, tcpdump) - CISSP (in progress)
- **AI & LLMs**: Self-hosted LLM/VLM & diffusion pipelines (Qwen, FLUX, Diffusers, Ollama) - LoRA training, identity conditioning (PuLID/IP-Adapter, face embeddings), inpainting, prompt compilation, agent memory, elastic GPU serving - Classical ML & computer vision (TensorFlow, OpenCV, CUDA) - Agentic coding & review with Claude, Codex, and Gemini
- **Embedded & Kernel Development**: Linux kernel modules & system calls - FreeBSD kernel modules - Low-level C/C++ (memory, concurrency, IPC)

## TECHNICAL AUTHORSHIP & OPEN SOURCE PROJECTS

- **The Software Ecology** (to be published, DeKrypt B.V.): Examines the software industry as a socio-technical ecology, where companies, teams, and regulators interact through human behavior, power, and organizational dynamics as much as technology.
- **Extreme C**: Book published October 2019 by Packt Publication.
- **Linux Two-Step Authentication with Rust**: liveProject for Manning Publications, with its accompanying ESS backend and client projects in Rust.
- **Other Rust projects**: Domain Catcher Engine and the Outlook .msg parser.

## FOREIGN LANGUAGES

Persian (**Native**), English (**Fluent**), German (**A1**), Dutch (**B1**), French (**A1**)

## EDUCATION

- (2017 - 2022) **Bachelor of Science in Physics (Astronomy and Planetary Science)** at The Open University
- (2004 - 2008) **Bachelor of Software Engineering** at University of Isfahan (Isfahan, Iran)

## PERSONAL INTERESTS

- **Physics and Astronomy**: Cosmology, Astronomy, Quantum Physics, Quantum Mechanics, Particle Physics, Special and General Relativity, Theory of Everything, Photonics.
- **Mathematics**: Linear and Abstract Algebra, Probability Theory, Information Theory
- **Cryptography**: Information Security, Network Security, Security Algorithms, Quantum Information Theory, Quantum Key Distribution, Quantum Cryptography
- **Philosophy**: Western Philosophy, Sociology, Economics, Politics and Humanities
- **Telecommunications**: Modulation, Multiplexing, Photonics, optical fibers, IEEE 802.3, Coding Theory